

Guest Editorial

Special Issue on Z-Pinch Plasmas

THIS special issue on Z-pinch plasmas in the IEEE TRANSACTIONS ON PLASMA SCIENCE is the seventh to be published [1]–[6], starting with the first one with Guest Editors J. Davis and C. Deeney in 1998. The publication of so many special issues in just 20 years is a testament to the continued vitality and variety of research in the area of pulsed-power-driven loads. The present Special Issue grew out of interest to record the 10th Dense Z-Pinch Conference held in Stateline, NV, USA, during August 2017. In addition to papers from attendees at this conference, the submission was open to other scientists who wanted to archive their research. As a consequence, this Special Issue has the largest number of published papers compared to previous ones. In addition to authors from the USA, there are articles from the U.K., Russia, China, Hungary, the Czech Republic, and Chile. The topics in this collection of Z-pinch research cover a broad range of load designs and applications: liners and foils; wire arrays; X-pinch; gas puffs; dense plasma foci (DPF); diagnostics; and devices and pulsed-power facilities—comprising many theoretical and simulation studies together with experimental findings. Most of the papers arise from work or study with university class drivers, ~ 1 MA and 100–200-ns implosion times, and there are a few reports of pinches on multi-mega-ampere (MA) generators.

With the recent development of the magnetized liner inertial fusion concept, there has arisen a strong interest in the detailed physics of liners and how they can be used to produce novel plasma configurations. Inverse liner loads expand outward and can produce strong shocks in a surrounding gas shell with a radiative precursor (Clayton *et al.*), but liners also develop structures on the surface at high current densities (Shelkovenko *et al.*). With the addition of an axial magnetic field in a Z-pinch liner, simulated Hall effects can play a role in the ablation process (Byvank *et al.*). The electrothermal instability, known to seed magnetohydrodynamic instabilities on pulsed-power-driven solid metallic targets, is studied by Steiner *et al.* using optical imaging of ablations from thin aluminum foils. The Nernst effect of magnetic field transport due to thermal gradients is another physical manifestation of the complexity in liner loads (Giuliani and Velikovich).

Wire array loads continue to be of interest in Z-pinch physics. It is noteworthy that planar wire arrays of tungsten can be imploded to produce soft and hard radiation on linear transformer drivers (LTDs) (Kantsyrev *et al.*). Axial diagnostics in wire array implosions present a different view of

the precursor dynamics (Ivanov *et al.*). Dynamic hohlraums have received considerable attention in past special issues on Z-pinch, and Chinese researchers are now modeling such loads for their 8-MA PTS driver (Ning and Chen). The combination of a Z-theta pinch with a wire array is calculated to produce ultrahigh pressures and soft X-rays (two papers by Selemir *et al.*).

X-pinch has long been used as back lighter sources and are known to produce high-energy electron beams. X-rays from an X-pinch indicate that explosive electron emission develops from a surface breakdown rather than follow the Ecton hypothesis (Pikuz *et al.*). Fast electrons in an X-pinch can lead to polarized emission lines as measured by Childers *et al.* X-ray bursts produced during collapse of an X-pinch continues to be of interest to understand energy balance and radiative properties of X-pinch. Pavez *et al.* present studies of such properties on a small, low-impedance, and capacitive generator. A variation of the X-pinch, namely, the hybrid X-pinch, can produce a localized spot of intense XUV radiation (Romanova *et al.*).

Another type of Z-pinch loads is a gas puff. Reversing the polarity of a gas puff can impact the radiation properties and enhance the influence of hot electrons on X-ray spectra (Shlyaptseva *et al.*). A recent innovation that increases the range of materials for this configuration is the metallic puff. In the presence of ambient axial magnetic field, the initial radius of the breakdown is found to change (Roussikh *et al.*). Gas puffs also have the capability to implode multiple shells or combine different gases for dynamical or diagnostics usage. A staged Z-pinch uses a heavy gas in an outer shell to implode on an interior gas target of lower atomic weight. Conti *et al.* characterize the injected heavy gas with interferometric and visible emission diagnostics in the staged pinch. A triple puff consists of two concentric outer and inner gas puffs and a central gas jet. Such loads provide a test bed for studies of shock heating and/or photoionization of the dense center jet by the imploding plasma and validating initial conditions for stable implosions (Qi *et al.*). Simulations of recent triple Ar puffs on the Z-generator with a small Xe dopant in the central jet show a marked decrease in the Ar K-shell yield in agreement with experimental data (Tangri *et al.*). DPF usually involve a static fill prior to current initiation. In an alternative design to mitigate insulator flashover, Beresnyak *et al.* simulate fast deuteron production for a DPF with three gas injectors downstream from the insulator stack. Another alternative use of a gas Z-pinch is an optical waveguide for a CO₂ laser pulse within a capillary Ar pinch, as modeled by Shaplov *et al.*

Advanced diagnostics typically lead to enhance understanding in experimental physics. Though proton deflectometry has been often used to study electromagnetic fields in laser-driven plasmas, Munzar *et al.* present the use of this diagnostic to measure the magnetic fields in a Z-pinch. The Thomson scattering diagnostic enables localized measurements of the plasma temperatures and densities. Work at Cornell University, Ithaca, NY, USA, has focused on time resolution of plasma jets (Banasek *et al.*) and gas puffs at implosion (Rocco *et al.*). A special design for a Thomson scattering measurement of a photoionized plasma on the 20-MA Z-generator at the Sandia National Laboratories, Albuquerque, NM, USA, is presented by Kozlowski *et al.* Finally, simulations to validate X-ray measurements imply the need to account for dead layers on filters (Repin *et al.*).

The last section collects papers that describe new devices and pulsed-power facilities. McBride *et al.*'s article is an invited tutorial on LTD generators employed at several research institutions for high-energy density physics. It was initially intended for a special issue containing lectures from the mini-course at the 2016 ICOPS meeting but has been included in this Special Issue because LTDs may be the future of Z-pinch experiments. Magnetic anvil cells use a gas, foam, or solid damper to compress a material sample via magnetic pinch forces. According to numerical calculations, they can produce warm dense matter and reach pressures on the order of 1 Mbar using an MA pulsed-power driver (Gourdain *et al.*). High-current discharges along a ferrite surface can follow a predefined initial discharge pattern and be used to generate and study intense, reproducible pulses of optical and XUV radiation (Tilikin *et al.*). Campbell *et al.* describe recent upgrades to the MAIZE LTD facility at the University of Michigan, Ann Arbor, MI, USA, and Taylor *et al.* present an overview of the Charger-1 pulsed-power facility at the University of Alabama, Tuscaloosa, AL, USA, for thermonuclear space propulsion.

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